OSI Layer

Tasks

1.Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

ANS:



No.	Layer Name	Example
7	Application	HTTP, WhatsApp
6	Presentation	JPEG, SSL
5	Session	NetBIOS
4	Transport	TCP, UDP
3	Network	Router, IP
2	Data Link	Switch
1	Physical	Cable, Hub

1. Application: Used to open websites and apps.
2. Presentation: Changes data into a readable format.

3. Session: Starts and ends communication.
4. Transport: Sends data safely.
5. Network: Finds the path for data.
6. Data Link: Transfers data between devices.
7. Physical: Sends data through cables and signals.

**2.Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.**
ANS:

1. a) Encrypting data: Presentation Layer (Layer 6) – Keeps data safe by hiding it.
2. b) Routing packets: Network Layer (Layer 3) – Finds the path for data.
3. c) Electrical signals: Physical Layer (Layer 1) – Sends data through cables.
4. d) Opening a website: Application Layer (Layer 7) – Helps us use websites.
5. e) Finding data errors: Data Link Layer (Layer 2) – Checks errors in data.

3.Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

ANS:
When I send a message on WhatsApp, different OSI layers work together.

1. Application Layer (Layer 7): I type and send a message.
2. Transport Layer (Layer 4): It helps send the message to the other person.
3. Network Layer (Layer 3): It finds the path to send the message.
4. Physical Layer (Layer 1): It sends data through Wi-Fi or mobile internet.

4.Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

ANS:

1. Application Layer (Layer 7): The website or browser may have a problem.
2. Transport Layer (Layer 4): HTTP/HTTPS connection may not be working.
3. Network Layer (Layer 3): Ping is working, so the IP connection is okay.